Binary Classification of AOI Alarms into True Defects and False Calls

Second check of AOI results with supervised machine learning (Siemens production data)

P2: core code and baseline simulation · test plan · report

Yu Han · Melisa Subasi · Meron Hagos · Alejandro Rivas

Team 1 · Guided by Prof. Dr.-Ing. Hamidreza Heidari · 3 November 2026 · github.com/bodhi584/aoi-false-call-classification

PROJECT PROGRESS

Where the project stands

P1 · 20 Oct — done
Proposal, tech stack, MVP (topic revised 23 Oct)

P2 · 3 Nov — this phase
Core code + baseline simulation; test plan; report

P3 · 24 Nov — next
Feature progress; unit tests and CI; issues board

P4 · 8 Dec — planned
Feature progress; full evaluation

P5 · 22 Dec — planned
Demo video, run instructions, packaging

Final · 12 Jan — planned
Final presentation and report

P2 in short

Literature · 18 sources. A machine-learning second check removes false calls on a test set, but on later production days it misses 5–15 % of true defects [7].

Our question · [7] names two causes, drift and an unstable threshold, and tests neither. Methods for both exist; they become the two variables of our simulation.

Data · Same public Siemens set as [7]. It changes over time: among flagged boards, real defects rose from 1–2 % per week to 3–6 % in the last 4 weeks.

Code · We reimplemented the method of [7] with our own code, then corrected the threshold and added retraining. Delivered as runnable Python code in the public repository, with its result files and figures.

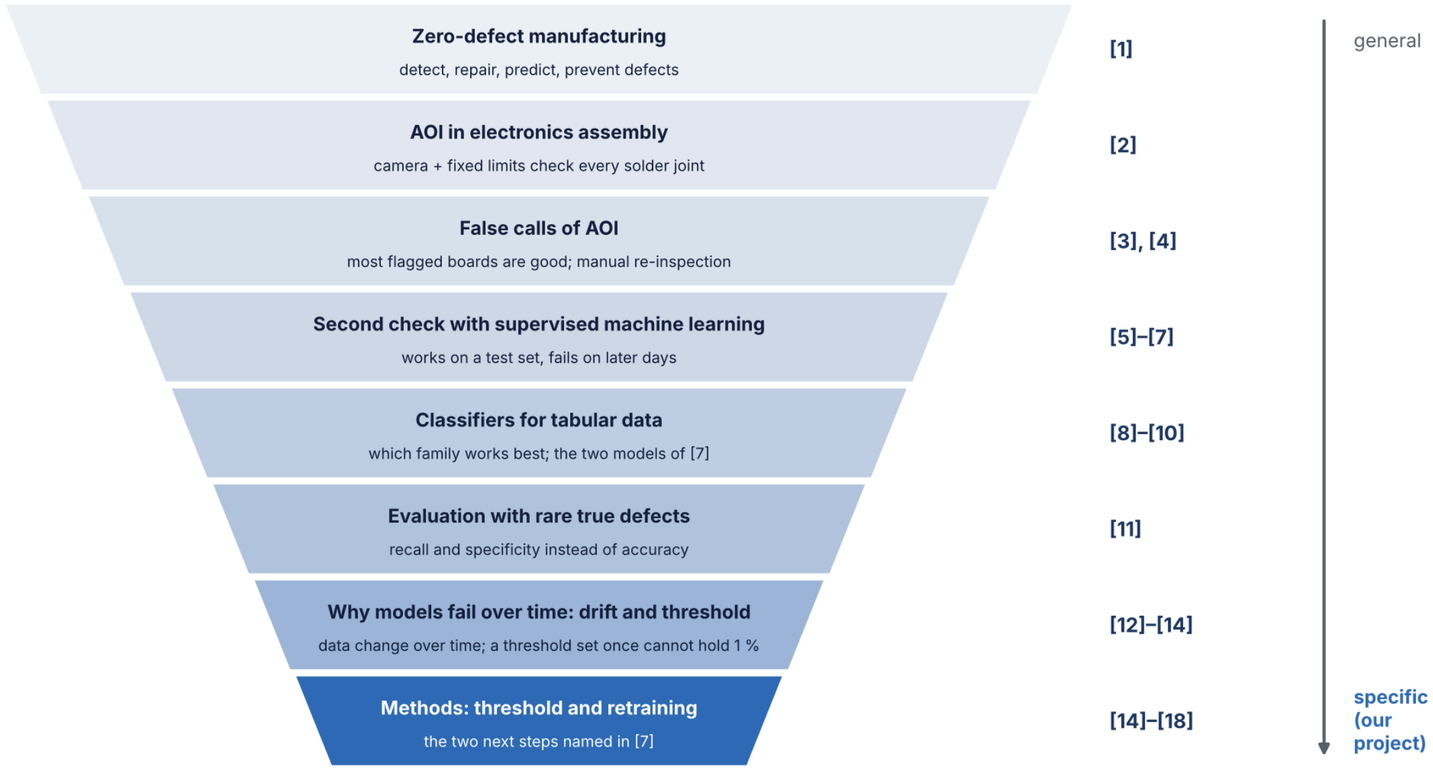
Baseline · Our code shows the same problem as [7]: 1–2 % missed true defects on the test set, 5–14 % on the 5 later periods. The two changes are tested in P3.

Test plan · DOE: threshold 3 · retraining 6 · retraining data 2 · model 2 → 72 settings × 10 seeds.

Next (P3) · Run the design of experiments: threshold first, then retraining. The simulation software follows in P4–P5.

Melisa Subasi

From general to specific: the literature in one picture



Our simulation: threshold and retraining strategies as variables, compared with [7].

The problem

≈ 99 % of the joints AOI flags are false calls; people re-check them all [4], [7].

Core paper [7]

Same data, public code. Trained once: ≈ 1 % missed on the test set, 5–15 % on later periods.

Bottom two levels

Two causes: drift and a threshold set once [12]–[14]. Methods for both: [14]–[18].

Melisa Subasi

Two causes named in [7] become the two variables of our simulation

Fails over time
5–15 % missed later [7]

→

Two causes
drift [12]; threshold set once [7], [14]

→

Methods exist
see table below

→

Work elsewhere
fraud [17], chip line [18]

→

Open for AOI
not yet tested

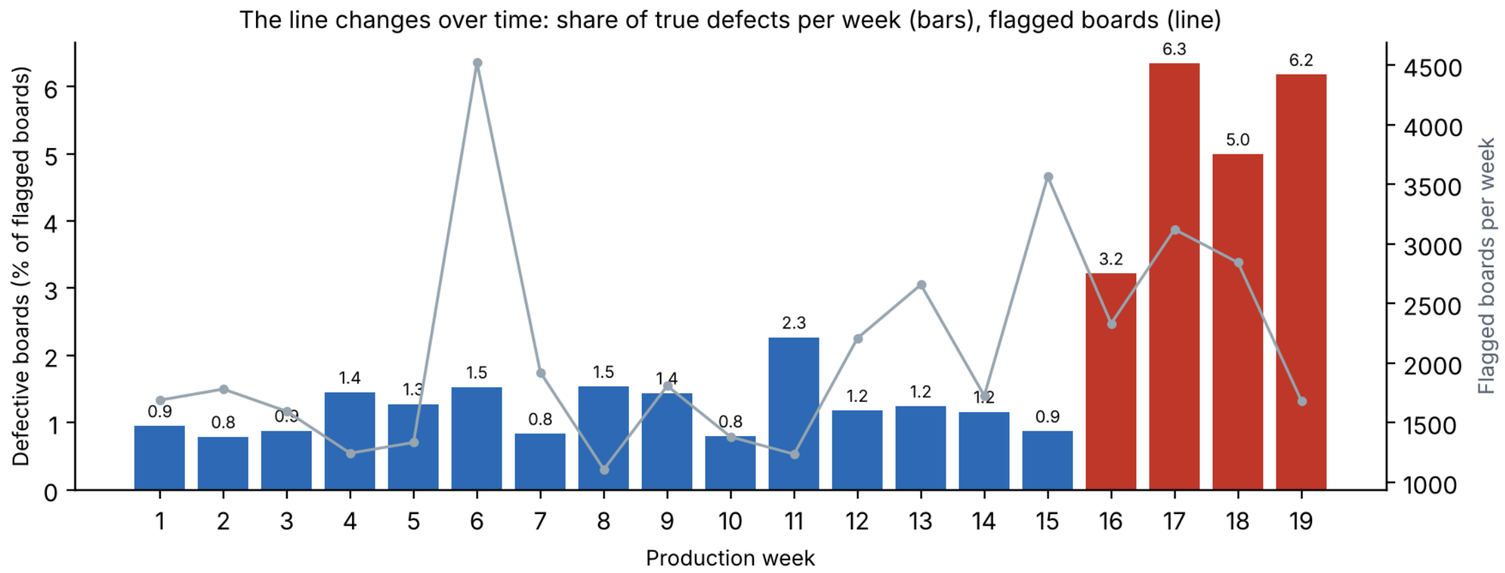
Variable	Option 1	Option 2	Option 3
Threshold	set once, as in [7]	safety margin for 1 % [14]	re-set per period [15]
Retraining	off, as in [7]	fixed schedule [12], [17], [18]	when drift is detected [16]

[7], Sec. VII: "… potential next steps to stabilize the modeling by improving the method for setting the threshold and handling the distribution drifts within the dataset."

Our simulation tests exactly these two steps.

Meron Hagos

The line changes over time



Facts (our analysis)

440,274 flagged joints, 39,742 boards, 132 days

97.8 % of flagged boards are false calls

Defective boards: 0.8–2.3 % per week, then 3.2–6.3 % in the last 4 weeks

Early and late days can be told apart (drift test, AUROC 0.974)

Meron Hagos

Input → system → output

Input = as [7]
Siemens AOI data, in time order [4]

→

System = our code (scripts/replicate_pfab2025.py)
[7] reimplemented + threshold corrected + retraining added

→

Output = as [7]
missed defects %, false calls released %

1 · Reimplemented

Own code from scratch: same split, models, settings from their logs, threshold method. Their code was read, not copied.

2 · Corrected

The threshold should come from the best of all training runs. [7] uses the threshold of the last run. We use the threshold of the best run.

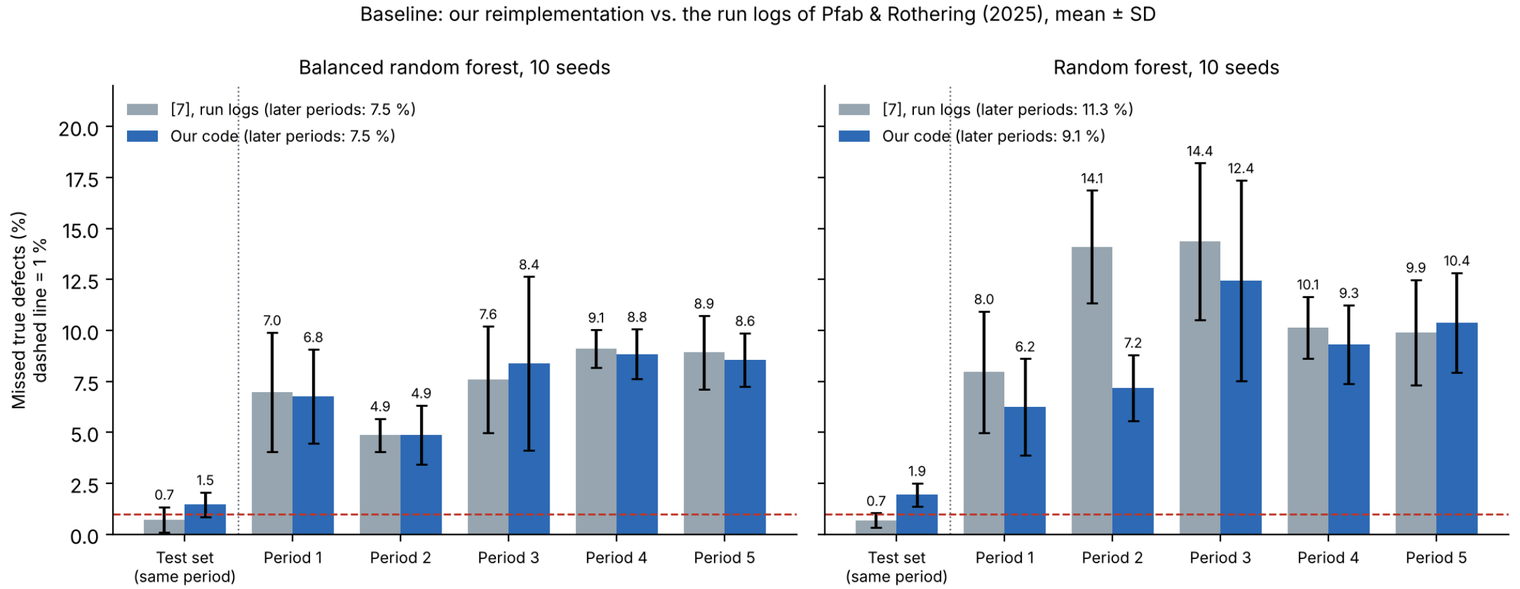
3 · Added

Retraining before each of the 5 later periods; [7] only trains once.

In the end, this system becomes a small web tool for the simulation: threshold and retraining are settings you can change.

Alejandro Rivas

Baseline: our code shows the same problem as [7]



Problem reproduced

Test set: 1–2 % missed. Later periods: 5–14 %.

Balanced RF

Later periods: 7.5 %, same as [7].

Small gaps, open

RF 9.1 % vs. 11.3 %; test set a bit higher. Checked in P3.

Alejandro Rivas

Factors, levels, constants, responses

Factor	n	Level 1	Level 2	Level 3	Level 4	Level 5	Level 6
A · Threshold	3	as in [7], corrected	safety margin [14]	re-set per period [15]	–	–	–
B · Retraining	6	never, as in [7]	per period	weekly	every 3 days	daily	on drift [16]
C · Retraining data	2	all data so far	recent window [12]	–	–	–	–
D · Model	2	random forest	balanced random forest	–	–	–	–

Held constant

Data and split, features, hyper-parameters of [7], recall target 99 %

Responses

Missed true defects (%) · false calls released (%), per later period

Design

3 × 6 × 2 × 2 = 72 settings, each with 10 seeds (replicates)

Yu Han

What changed since P1, and what comes next

	P1 plan (23 Oct)	Now, after finding [7]
Second check	build our own	reimplement [7] as the baseline
Models	logistic regression, RF, neural net	RF, balanced RF, as in [7]
Measures	recall ≥ 99 %, specificity ≥ 20 % (boards)	missed defects, false calls released, as in [7]
Focus	day-by-day simulation	threshold and retraining, the two next steps of [7]

P3 · 24 Nov

Run the DOE (slide 8): threshold first, then retraining. Unit tests on the core code.

P4 · 8 Dec

Full evaluation; simulation software with threshold and retraining as settings.

P5 · 22 Dec

Packaging, run instructions, demo video.

Repository

github.com/bodhi584/aoi-false-call-classification

Thank you

Team 1 · Yu Han · Melisa Subasi · Meron Hagos · Alejandro Rivas

The problem and the second check

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[3] Siemens, “Siemens and NVIDIA advance AI-powered PCB inspection from Digital Twin to factory,” press release, Plano, TX, USA, Oct. 21, 2026. open

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[7] K. Pfab and M. Rothering, “Towards improved research methodologies for industrial AI: a case study of false call reduction,” arXiv:2506.14521, 2025. Code and run logs: gitlab.com/research9841542/FCR_1. (Core paper) open

Methods for drift and the threshold

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[17] A. Dal Pozzolo, G. Boracchi, O. Caelen, C. Alippi, and G. Bontempi, “Credit card fraud detection: a realistic modeling and a novel learning strategy,” IEEE Trans. Neural Networks and Learning Systems, vol. 29, no. 8, pp. 3784–3797, 2018. open

[18] M. Gao, J. M. Perathoner, A. L. Bonin, S. Eulig, and G. Klesse, “Robust and reliable AI for predictive quality in semiconductor materials manufacturing with MLOps and uncertainty quantification,” arXiv:2605.07752, 2026. open